

# JASON HON

SOPHMORE COMPUTER SCIENCE MAJOR

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## ⚙ ABOUT ME

**Languages** | Java, Python, C, C++, Javascript, Typescript, Scheme, Bash, CSS, HTML, SQL

**Technologies** | EC2, Flask, Express, MongoDB, Node, React, Angular, Tailwind

**Development** | Git, Linux, Make, Valgrind, Address Sanitizer, GDB, GCC

## 🏗 PROJECTS AND OTHER EXPERIENCES

### 🔗 SLMod — Java, Flask, MongoDB

- Produced a portable **Java** QOL game modification used by over **40** users daily at its peak
- Led a team of 3 developers** to develop a **Flask** user-tracking system, data stored using **MongoDB**
- Visualized **over 6,800** user log-ins on an online dashboard, implementing **OAuth2** for secure access.

### 🔗 Chess Engine — C++

- Developed a Chess engine in C++ featuring various levels of **AI opponents**, using advanced decision-making algorithms to incorporate piece values and to calculate complex evaluations
- Strictly adhered to **OOP** principles and advanced design patterns to create a structured and maintainable codebase, ensuring high cohesion and low coupling
- Led a team of 3 to maintain comprehensive documentation and UML diagrams to streamline development

### 🔗 Portfolio Website — React

- Developed a modern and responsive portfolio website using **React**, showcasing past experiences.
- Utilized React Router for seamless navigation between different sections of the website

### 🔗 UW Assignment Planner — MongoDB, Express, Angular, NodeJS

- Created an assignment planner for UWaterloo students using the **MEAN** stack
- Built **RESTful APIs** with **Express.js** for storing and updating tasks
- Integrated third-party APIs** (e.g. Google calendar and notification) to enhance functionality

### 🔗 Boulder Solver — Java

- Developed a solver that solves complex boulder puzzles (Sokuban) in **Java** in requirements of a client
- Engineered a backtracking algorithm that generates the most optimized path for any maps in under **100ms**

## 🏢 WORK EXPERIENCE

### Bonumcare — CSS, HTML, and JavaScript

Software Developer, Remote

Seattle, WA (Summer 2023)

- Developed a **GPT-3.5** based "Conversation Buddy" to chat with lonely elderly individuals
- Implemented APIs on an **Express** server to send and receive over 300 messages daily
- Maintained and updated the chatbot's knowledge base, ensuring the accuracy and relevance of information in its output

### Content Creation

YouTube

(March 2022 - May 2023)

- Produced video content for a broad audience of an average of **60,000** viewers on YouTube
- Negotiated and finalized strategic partnerships with multiple sponsors (e.g. **Plarium**, **Akrew**)

## 🎓 EDUCATION

### University of Waterloo

Waterloo, ON

Honours Bachelor of Computer Science, Artificial Intelligence Specialization

- GPA: 3.95 / 4.0 (CAV: 92.1%)
- Relevant Coursework: Data Structures, Object Oriented Programming, Algorithm and Data Abstraction, Compilers, Tools and Techniques for Software Development
- Honours: CS Intl. Student Upper-Year Scholarship, Presidential Scholarship of Distinction, Term Distinction (Totalled over CAD\$ 8,000)
- Activities and Societies: Computer Science Club, Data Science Club, Chinese Student Association